Game title
Quantum Guardians: SDG4(TBD)
Developer(s)
Aleksandre Titberidze at23025

Core Concept
Quantum Guardians: SDG4 is a turn-based pixel RPG where the "enemies" are not monsters in a traditional sense, but personified social challenges that harm education systems—dropout, burnout, misinformation, bureaucracy, and inequality. The player controls Inspector and fights these "crisis entities" to restore progress toward **SDG 4: Quality Education**.
The experience aims to feel like a retro JRPG battle system (simple animations, clear turns, skills, HP), but with a unique strategic layer: outcomes are shaped through **quantum-inspired mechanics** such as interference, entanglement, and measurement/collapse. The player must decide not only which skills to use, but also when to "measure" the situation for certainty—because observation has consequences.

Thematic Focus
The main theme is **Social Good through SDG 4 (Quality Education)**. The game frames education as a system affected by real-world pressures: unequal access, teaching overload, poor infrastructure, and policy distortions.
The theme is integrated directly into gameplay through battle objectives and mechanics:

- Enemies represent education-related crises (The sprites which are at the bottom but the names will be related to crises.).
- Player actions represent interventions (support programs, equity actions, wellbeing breaks, formative assessment).
- The "damage system" is tied to education dimensions: **Access**, **Quality**, and **Equity**. The player wins fights by improving these dimensions rather than simply "attacking for HP."

This means social good isn't just a story layer; it is the functional model that drives battle outcomes, progression, and failure conditions.

Quantum Dimension
Quantum concepts used

- **Superposition**: enemy weaknesses and/or system states start uncertain (e.g., crisis is not clearly weak to Access vs Quality vs Equity).
- **Measurement**: "Audit/Exam/Survey" actions reveal (collapse) the hidden weakness/state.
- **Collapse**: once measured, the state becomes fixed/known for a period.
- **Interference** (core): player actions contribute signed effects that can amplify or cancel each other (constructive vs destructive interference).
- **Entanglement**: certain enemies or stats can be linked, so changes to one propagate to another.
- **Observer effect**: frequent measurement increases **Pressure/Burnout**, creating long-term penalties (teaching-to-the-test, stress, rigidity).

Level of implementation
Mechanical + state-based implementation (not just conceptual).
The game does not implement full quantum gates, but it implements core quantum behaviors through:

- explicit internal "state" variables (uncertainty flags, entanglement links, measured/collapsed state timers)
- interference scoring (nonlinear combination with cancellation)
- measurement changing the system (pressure increases, future outcomes shift)

How it goes beyond classical probabilistic randomness

A purely classical model could represent uncertainty as "random chance." This design goes beyond that by including **structure**:

- **Interference**: effects are not additive; two actions can cancel each other out (destructive interference), or amplify (constructive interference). This is not equivalent to "more/less probability," because *sign and order* matter.
- **Entanglement**: outcomes across targets are correlated through explicit linking rules, not independent random rolls.
- **Measurement changes the system**: observing does not only reveal information; it modifies future potential via Pressure/Burnout penalties.

These features ensure the model is not reducible to standard "RNG damage" mechanics.

Game Mechanics and Rules

Turn structure (turn-based)

A battle runs in rounds. Each round:

1. **Player Phase**
    - Player selects one action per hero (attack/support/measure/entangle).
2. **Resolution Phase**
    - Actions apply amplitudes and state effects.
    - Interference and entanglement rules update results.
3. **Enemy Phase**
    - Enemies apply crisis effects (increase burnout, reduce access, flip phase, entangle, etc.).
4. **End Phase**
    - Status durations tick down (measured state, entanglement duration, debuffs).

Player actions and choices

Player skills are built around three education dimensions:

- **Access actions (A):** reduce dropout pressure, improve attendance effects.
- **Quality actions (Q):** improve learning outcomes, pedagogy, teacher development.
- **Equity actions (E):** reduce inequality, target support to at-risk groups.

Plus "quantum tools":

- **Measurement (Audit/Exam):** reveals weakness/state but increases Pressure/Burnout.
- **Entangle:** link two entities so a portion of effects transfers between them.
- **Phase/Interference tools:** flip a sign/phase of the next action, or protect from cancellation.

System responses

- Enemies respond with mechanics tied to their theme:
    - **Dropout Goblin:** attacks Access and hides weakness until measured.
    - **Burnout Orc Teacher:** punishes over-measurement; increases Pressure fast.
    - **Skeleton Politician Knight:** introduces bureaucracy/decoherence-like penalties (reduced effectiveness, delayed actions).

Constraints and limitations

- **Energy/Action limits** per turn (simple 1 action per hero).
- **Pressure/Burnout meter**: rises with measurement and some enemy attacks.
- If Pressure reaches a threshold, the party receives debuffs (reduced Quality effect, more cancellation, higher failure risk).

Narrative / Setting
The game uses a symbolic fantasy setting where a society's "Education Realm" is threatened by crisis entities. The heroes represent forces of improvement: teaching, community support, and ethical governance. Battles represent real-world challenges framed through fantasy metaphors.
The narrative supports theme by:

- Naming enemies after education problems (dropout, burnout, bureaucracy).
- Keeping the player's objective aligned with social good (restore learning, equity, access).
- Making "measurement pressure" part of the story (schools adapt to scrutiny; wellbeing matters).

Winning and Losing Conditions
Winning (battle-level)
Win a battle by defeating the crisis entity, defined as:

- reducing the enemy's "Crisis HP" to zero through effective A/Q/E actions, **or**
- meeting a battle objective (e.g., stabilize Access above a threshold within N turns).

Losing (battle-level)
Lose if:

- party HP reaches zero, **or**
- **Pressure/Burnout** exceeds the maximum threshold (system collapse / strike / unrest).

Campaign endings (prototype-friendly)

- **Good ending:** balanced success across Access, Quality, Equity with manageable burnout.
- **Mixed ending:** success in one dimension but failure in another (e.g., high Quality but high inequality).
- **Bad ending:** burnout/pressure collapse or repeated failures.

Platform and Tools

- **Engine:** Godot (2D)
- **Language:** GDScript
- **Assets:** 32×32 pixel spritesheets (idle/attack/hit/death), simple UI, SDG icons as menu/chapter screens
- **Planned technical scope: Playable prototype**
    - 1 chapter (SDG4)
    - 3 enemies + 1 boss (or 3 elites)
    - 1 playable character(start)
    - 10–15 skills total
    - Quantum features fully working: interference + entanglement + measurement/pressure

Assets for enemies

Assets for enemies plus playable characters.(if there will be multiple)